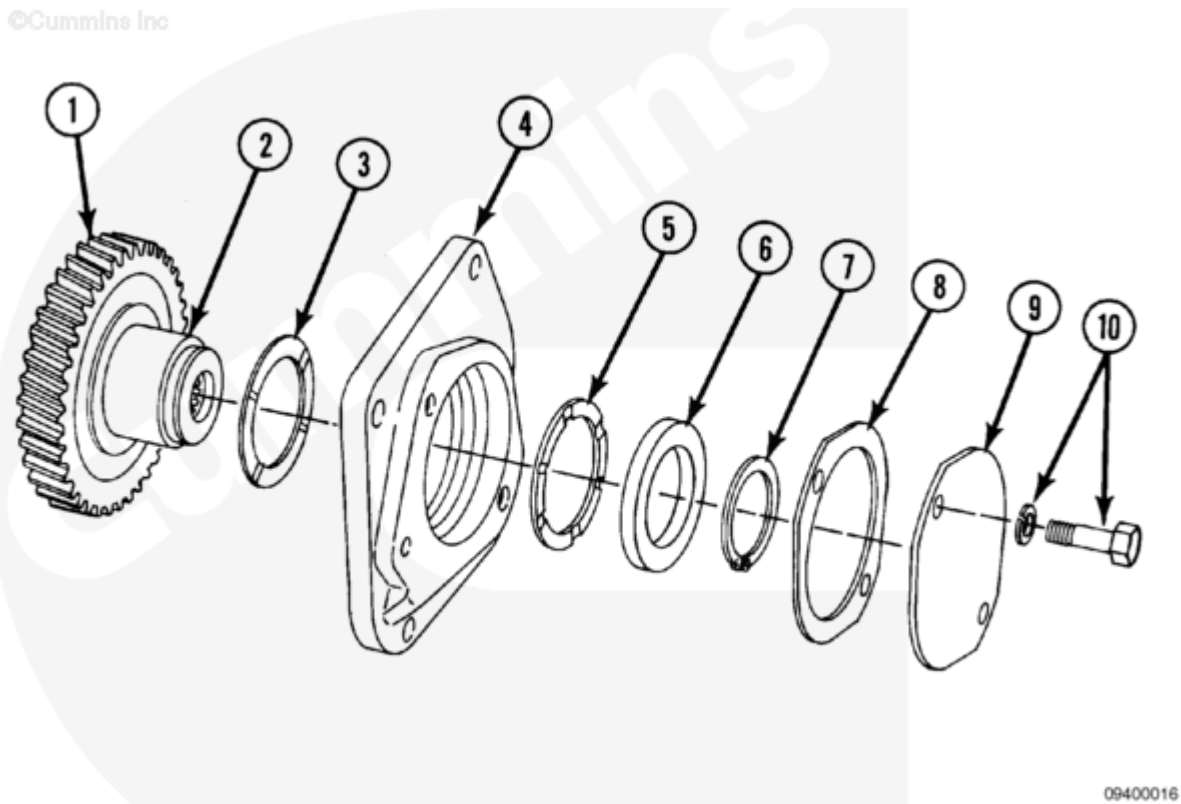


Trainings ▾

Exploded View



1. Hydraulic pump drive gear
2. Hydraulic pump drive shaft
3. Thrust bearing
4. Hydraulic pump drive housing
5. Thrust bearing
6. Clamping washer
7. Retaining ring
8. Gasket
9. Hydraulic pump cover
10. Lock washer and capscrew.

Preparatory Steps

Note : Some engines contain a cover plate instead of a hydraulic pump.

- Remove the hydraulic pump. See equipment manufacturer service information.



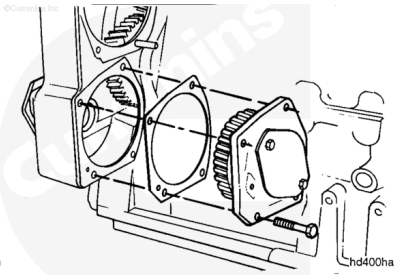
ck800wa

Remove

Remove the four hydraulic pump drive mounting capscrews.

Remove the hydraulic pump drive.

Remove and discard the gasket.



hd400ha

Inspect for Reuse

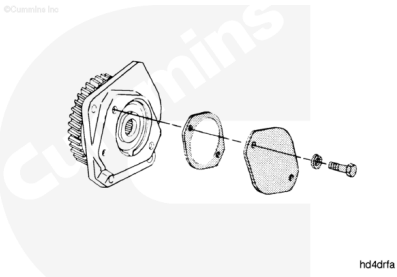
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Clean the hydraulic pump drive housing and gear with solvent.

Remove the cover from the pump.

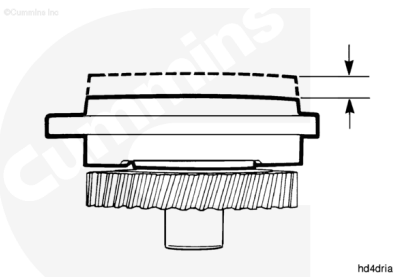
Remove and discard the gasket.



hd4drfa

Measure the hydraulic pump drive end clearance.

Hydraulic Pump Drive End Clearance		
mm		in
0.13	MIN	0.005
0.48	MAX	0.019



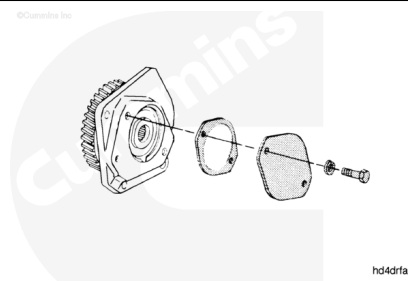
hd4drfa

If the hydraulic pump drive is **not** within specifications, it **must** be reconditioned.

Install a new gasket.

Place the cover onto the pump drive.

Install the lock washers and capscrews.

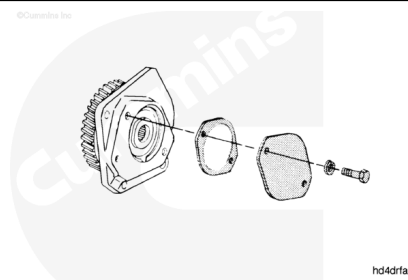


Disassemble

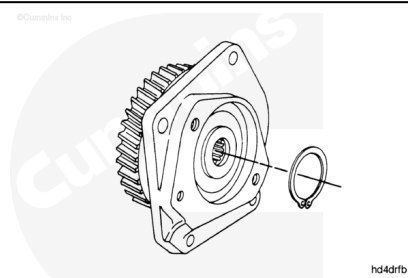
Remove the capscrews and washers.

Remove the cover.

Remove and discard the gasket.

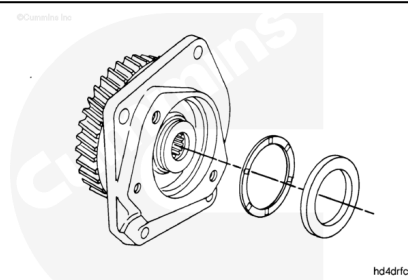


Remove the retaining ring.



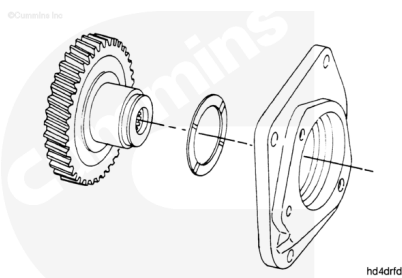
Remove the clamping washer.

Remove the thrust bearing.



Remove the shaft and gear assembly from the housing.

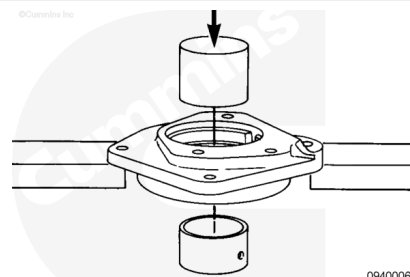
Remove the thrust bearing.



Note : Only replace the bushing if it is damaged or is not within specifications.

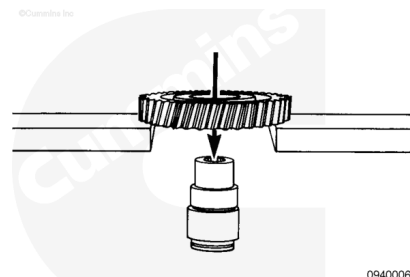
Support the housing in an arbor press.

Press the bushing out of the housing with a mandrel and arbor press.



Note : Only remove the gear from the shaft when the gear or the shaft must be replaced.

Remove the gear from the shaft with an arbor press.

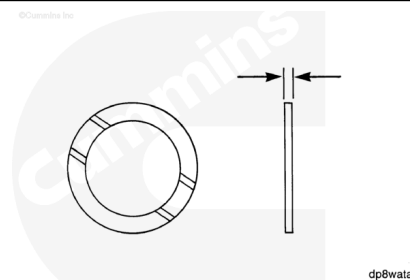


Clean and Inspect for Reuse

Check the grooved surfaces of the bearing for damage.

Measure the thrust bearing thickness.

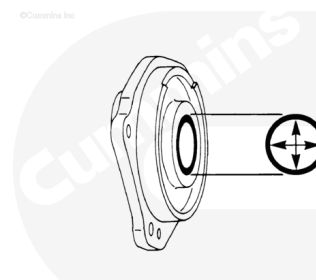
Thrust Bearing Thickness		
mm		in
2.286	MIN	0.090
2.311	MAX	0.091



If the bearing is damaged or **not** within specifications, it **must** be replaced.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



hd4hsta

Clean the housing with solvent.

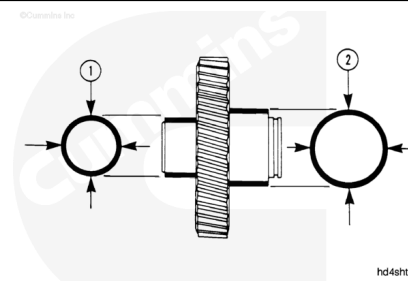
Measure the bushing inside diameter.

Bushing Inside Diameter		
mm		in
50.813	MIN	2.001
50.902	MAX	2.004

If the bushing is **not** within specifications, it **must** be replaced.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



hd4shta

Clean the shaft and gear with solvent.

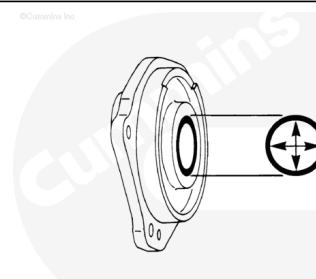
Measure the shaft outside diameter.

Shaft Outside Diameter			
	mm		in
(1)	38.062	MIN	1.4985
	38.072	MAX	1.4990
(2)	50.749	MIN	1.9980
	50.762	MAX	1.9985

If the shaft is **not** within specifications, it **must** be replaced.

Measure the housing bushing bore inside diameter.

Bushing Bore Inside Diameter		
mm		in
54.86	MIN	2.160
54.89	MAX	2.161

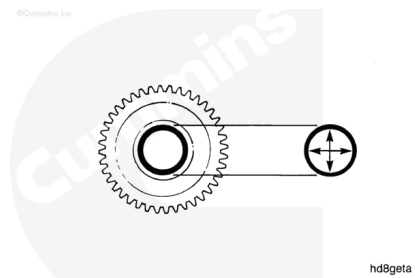


hd4hsta

If the bushing bore is **not** within specifications, the housing **must** be replaced.

Measure the gear inside diameter.

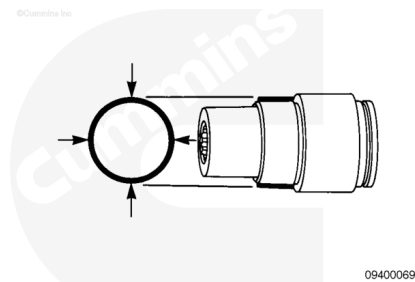
Gear Inside Diameter		
mm		in
44.399	MIN	1.748
44.425	MAX	1.749



If the gear is **not** within specifications, it **must** be replaced.

Measure the shaft diameter at the gear location.

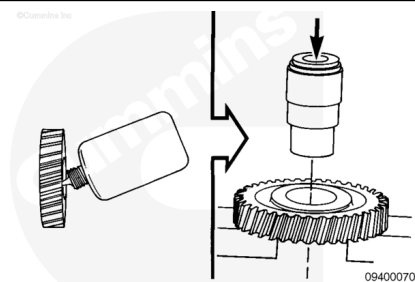
Shaft Outside Diameter at Gear Location		
mm		in
44.450	MIN	1.7500
44.453	MAX	1.7501



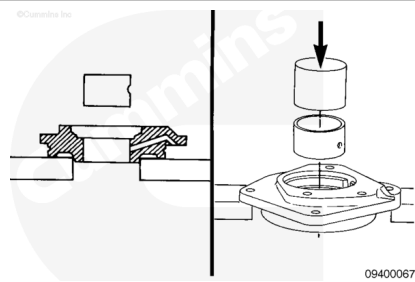
If the shaft is **not** within specifications, it **must** be replaced.

Assemble

- Apply a smooth coating of Loctite® 609, or equivalent to the inside diameter of the gear.
- Support the gear in an arbor press with the part number down.
- Press the shaft through the gear until the shoulder of the shaft touches the gear.



- Place the housing in an arbor press.
- Align the oil hole in the bushing with the oil drilling in the shaft.
- Press the bushing into the housing with a mandrel and arbor press.
- Both ends of the bushing **must** be positioned even with, or below, the surface of the housing.

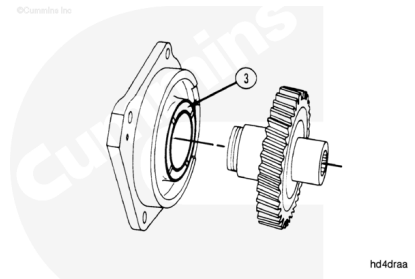


Lubricate the grooved surface of the thrust bearing with Lubriplate™ 105 or equivalent.

Position the grooved surface of the thrust bearing (3) as illustrated in the graphic.

Slide the gear and shaft assembly into the housing.

The gear **must** touch the thrust bearing.

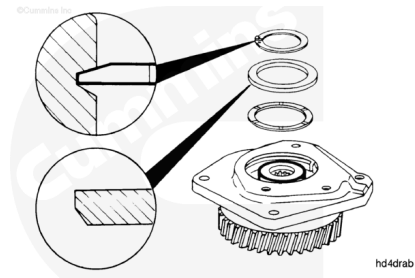


Lubricate the grooved surface of the thrust bearing with Lubriplate™ 105 or equivalent.

Slide the thrust bearing over the shaft with the grooved side positioned up.

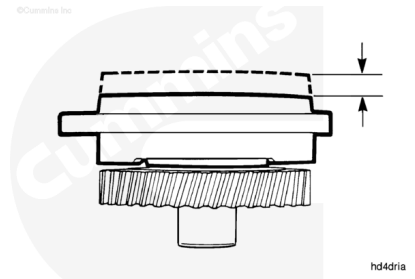
Slide the clamping washer over the shaft with the beveled edge positioned against the thrust bearing.

Install the retaining ring with the beveled edge positioned as illustrated in the graphic.



Measure the hydraulic pump drive end clearance.

Hydraulic Pump Drive End Clearance		
mm		in
0.13	MIN	0.005
0.48	MAX	0.019

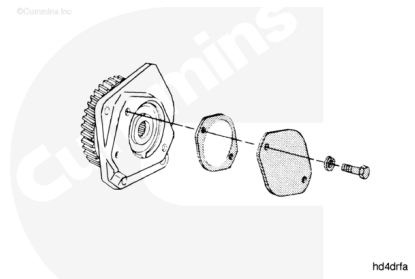


If the hydraulic pump drive is **not** within specifications, it **must** be reconditioned.

Install a new gasket.

Install the cover.

Install the lock washers and capscrews.



Install

The hydraulic pump drive or cover plate from engines with a two-piece front cover **must** have an o-ring in addition to the gasket.

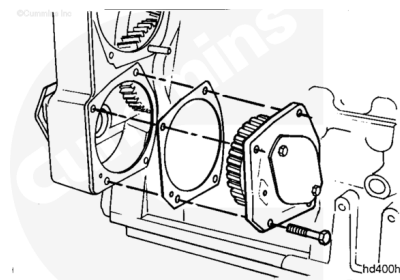
Lubricate the bushing in the front gear cover with clean engine oil.

Install the gasket, hydraulic pump drive, lock washers, and capscrews.

Tighten the capscrews.

Torque Value: 48 n•m [35 ft-lb]

Use the torque value above to tighten the capscrews a second time.



Finishing Steps

- Install the hydraulic pump. See equipment manufacturer service information.

